

LIBRARY MANAGEMENT SYSTEM

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By: Eunice Borres

Student ID: 922400749

Github: eunmdb

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Product Description

The Library Management System's goal is to create an efficient and effective way of managing various library systems, such as keeping track of borrowed books, basic information about patrons, a catalog of all the books available from different library branches, and much more. By using our management system, both employees and patrons will be able to have access to their library's extensive collection and accomplish various tasks online instead of having to go to the closest library branch. One key feature that our management system focuses on is the holds on library books. Some books tend to become more popular than others, resulting in high demand. However, due to the limited availability, not all patrons will have the ability to borrow and read the book. To resolve this issue, we plan on introducing a first come, first serve hold system to allow patrons to place a reservation on books that are currently being borrowed by their fellow patrons. This system would also keep track of patrons who have accessibility issues and would allow their books to be delivered instead of them having to pick up the books from their local library. Our database system will also store digital libraries such as audiobooks and research papers for those who need them. The database will also be designed in a way for the employees to view information regarding the patrons when they visit the library. Information such as overdue fines, near-due books, and preferred contact methods are displayed to easily help the employees assist their patrons.

Use Cases

1. Use Case: Limited availability of popular books

Actor: Michelle (Librarian), Joshua (Library Patron)

Description: The use case begins with Michelle who is a librarian at the Main Library.

She's currently assisting patrons with borrowing books, one of which is this season's most popular book. Joshua, a frequent patron of the library, also wants to borrow the said book.

Unfortunately for him, Michelle lets him know that there are no more copies available and that he will have to come back some other time when the books have been returned.

However, Michelle lets Joshua know that there is a way for him to place a reservation on the book so that he would be informed when a copy has been returned to the library and will be available for him to borrow, to which Joshua agrees. Michelle uses the system to access Joshua's profile and puts a reminder that he has a reservation on the next book that will be returned. The system also has a way to prevent patrons from reserving too many books at a time. If the patron has more than ten book reservations, the system will ask the patron to either cancel one of their reservations or to reserve the book later after they have fewer than ten reservations.

2. Use Case: Patrons that needs better accessibility

Actor: Sophia (Library Patron)

Description: Sophia loves going to her local library and looks forward to going there every week to browse the newly arrived books. However, due to an injury, she was required to be homebound and would be required to use a wheelchair for the next few months until her feet recovers. Devastated that she will not be able to go to her local library, she uses the library's website and places an order for the books that she wishes to read and have them delivered to her house. To return the books, she will use the same packaging that the books came in and would ask someone to drop them off at the local post office. To make sure that all the books would get returned, each person is allowed to have ten books delivered to them, afterwards they would have to return a book if they wish to borrow more.

3. Use Case: Overdue books

Actor: Jennifer (Librarian)

Description: Jennifer is a librarian at the Main Library branch. She notices that there are books that have been borrowed for months and have yet to be returned. The library has a policy that states that books can be borrowed for two weeks and would then have to be returned to the library, or there would be a 10-cent fine per day that it was not returned. Jennifer now needs to find a way to find patrons who have overdue books and send them an email notification regarding the books and payment. She uses the library's database system to search for these patrons and sends them an invoice and reminders regarding their overdue books. The system also allows her to send a message to the patrons who have a hold on the books to kindly wait until they have more available copies of the book. Patrons who have not paid or returned their overdue books will be prohibited from borrowing another book until their fine is paid and the book is returned if it was not declared as lost.

4. Use Case: Catalog

Actor: Julia (Library Manager)

Description: Julia is currently the library manager at her branch. With the new year arriving, her branch receives a shipment of newly published books, waiting to be borrowed by their patrons. She needed a way to keep track of the newly acquired books, as well as a way for her patrons to know the books that the library currently has. She uses the library's database system to update its collection of books and once all the information regarding the books has been entered, she publicizes this catalog for the patrons to use. This allows the patrons to search through their catalog, as well as identify which books are available at certain branches. To make it easier for both the patrons and the librarians to figure out which book they currently have, the catalog has filters that can be used such as the author's last name, alphabetical search, ISBN, genre as well as the year of publication.

5. Use Case: Digital Library

Actor: William (Library Patron)

Description: William is a frequent library goer but has terrible vision even when he is wearing glasses. His favorite author has finally released a book after five years and is excited to borrow it from his local library. After borrowing the physical copy of the book, he finds that the print is too small for him to read and disappointingly returns the book to his local library. The local librarian informs him that he can use his library card and the library app to borrow the digital copy of the book instead, which has features such as changing the font sizes. Similar to physical copies being borrowed, each patron can borrow up to ten digital copies at a time, and they would have to return the borrowed digital copies if they wish to borrow more. To prevent copyright issues, copy-pasting, and screenshots would not be allowed inside the app.

Functional Database Requirements

1. General User

- 1.1. Each user can only create one account using one email
- 1.2. Every user who makes an account will receive a library card
- 1.3. Every general user can use the library's system to browse for items
- 1.4. Any general user can view the upcoming events happening at different library branches
- 1.5. Any user can view the library's current books of the month of the library's website
- 1.6. Any user can join library events but for events with limited capacity, library patrons will have first priority
- 1.7. Every user can use the library's wifi even if they are not a library patron
- 1.8. Children under the age of 18 must be accompanied by a guardian or an adult to sign up for the library

2. Registered Use

- 2.1. Each user can checkout at most ten physical or digital books
- 2.2. Each user can create a hold on books for at most ten items
- 2.3. Each user must pay for any overdue books before being able to borrow new books
- 2.4. Every user will have a unique library card number which will be connected to their account
- 2.5. A library patron can request up to 10 books to be delivered to their house, if they have no outstanding fees
- 2.6. Each user is able to pay off their fees online, or at the nearest library branch

- 2.7. Many users can place a hold for any books
- 2.8. Every user can suggest a book for the library to buy with a limit of 10 books per month
- 2.9. Every user will have a specific pin that they will set up that is associated with their library card number
- 2.10. Users must renew their library card every 4 years
- 2.11. Library patrons under the age of 18 will not be fined for overdue or missing books. Instead, the adult who accompanied them to register will be fined instead
- 3. Books
 - 3.1. A book can be checked out by many users
 - 3.2. Each book can be placed on hold at most by 20 users
 - 3.3. Each book can be borrowed for at most 2 weeks
 - 3.4. Every book contains a special ID number identifying
 - 3.5. The library will have five copies of each book
 - 3.6. Borrowed books that are near their due dates can be extended for 1 week for 3 times if there are no holds on the books and if the account does not have any outstanding fee
 - 3.7. Extra large print items (books that are the size of an encyclopedia) cannot be taken outside of the library
- 4. Catalog
 - 4.1. Every book will have the name of the author, title and genre stored in the catalog
 - 4.2. Every book can be sorted or searched using the author's last name, title, genre and series if possible

- 4.3. The catalog can also be sorted by currently available books, along with the location of the branches that the book is available in.
- 4.4. Books that are written in other languages will be searchable using the title in the original languages, as well as the translated English title
- 5. Employees
 - 5.1. Every librarian can make book holds for a patron
 - 5.2. Every librarian can send emails to patrons reminding them of their overdue books
 - 5.3. Each employee will have a specific role
 - 5.4. Every library will have a library manager which is also a librarian
 - 5.5. Each library employee shall be paid in an hourly basis
 - 5.6. Each librarian is assigned to work in at least one library branch
 - 5.7. Every librarian is also considered as a library patron and can use their employee IDs to borrow books or digital items
 - 5.8. Each librarian is able to send notifications to patrons regarding their fees
- 6. Digital Library
 - 6.1. A digital item can be borrowed for two weeks
 - 6.2. Copy-pasting is not allowed for any digital items
 - 6.3. Font-sizes as well as color schemes can be modified to any digital item
 - 6.4. An designated app will be used to access the library's digital content
 - 6.5. Users will need to sign in to the app every 6 months to ensure that the credentials are still valid
 - 6.6. Digital items can be downloaded and saved offline only inside of the app

- 6.7. The app shall be able to load and retrieve saved offline items even if there is no internet connection
- 7. Universities
 - 7.1. All students and faculty staff will have an account
 - 7.2. All students and faculty staff will use their ID to log in to their accounts
 - 7.3. IDs will serve as the library card that will be used to scan or swipe when checking out items
 - 7.4. Laptops or tablets can be borrowed by any student that needs it and can be taken outside of the library but must be returned at the end of the semester
 - 7.5. Students who did not return the laptop or tablet at the end of the semester will be fined and the amount will be reflected on their student balance
 - 7.6. Digital research papers can be accessed from the school's website using the student ID
 - 7.7. Every student can reserve a study room inside the library for a maximum of 2 hours per day
 - 7.8. Scholarly articles as well as peer reviewed articles can be found in the university's library database
 - 7.9. Students may send in 10 requests per semester for a research that is not yet purchased by the university
- 8. Other Materials
 - 8.1. Newspapers, theses, encyclopedias and reference items cannot be borrowed or taken outside of the library

- 8.2. DVDs, CDs and other multimedia devices can be borrowed by any registered user for 1 week
- 8.3. Multimedia items that are near their due date can also be extended for 1 week up to 3 times if there are no holds and the account has no outstanding fee
- 8.4. Laptops and tablets can be borrowed by any patron for a maximum of two hours per day but cannot be taken outside the library
- 8.5. Each library patron can print at most 10 black and white pages and 5 colored pages per day. Any additional printing requires a payment of 10 cents per page for black and white and 40 cents for colored printing
- 8.6. Partnerships with other websites such as learning websites can be accessed using the patron's library card number and pin number.

Non-functional Requirements

1. Security
 - 1.1. All pins and passwords and user information shall be hashed and encrypted
 - 1.2. Pins shall contain a 5 numbers and passwords shall contain at least 8 characters with a mixture of lower case, uppercase and numbers
 - 1.3. The database shall automatically be backed up at 11:59pm
 - 1.4. All users will have the option for signing up for a two-factor authentication to secure their account
 - 1.5. Every unknown login location will require a two-factor authentication sent to the user's email to verify their identity
2. Performance
 - 2.1. Queries for data shall be returned in at least 5ms
 - 2.2. More specific queries shall be returned in less than 10ms
 - 2.3. The database should be able to handle at least 100 requests at a time
3. Scalability
 - 3.1. The system shall be able to support at least 50 concurrent users at a time
 - 3.2. During standard network speed, the website and app shall be able to load in less than 10 seconds
4. Maintainability
 - 4.1. After a system failure, the database shall not take more than 3 hours to recover
5. Memory Usage
 - 5.1. Under extreme usage and stress, the database shall not consume more than 60% of the memory

6. Storage Capacity

6.1. Each table shall be assigned 10mb of memory

6.2. The maximum amount of memory that each table can take is 2gb

7. DBMS

7.1. Separate tables will be used to store library patrons, employees, books, and articles